

Fruit fly assay report

Session: /home/team8/z.avi_assay_tests/assayfinal/outputs/assay/assay_20260414_122649

Run dir: /home/team8/z.avi_assay_tests/assayfinal/outputs/assay/assay_20260414_122649

Raw video:

/home/team8/z.avi_assay_tests/assayfinal/outputs/assay/assay_20260414_122649/raw_video.mp4

Background:

/home/team8/z.avi_assay_tests/assayfinal/outputs/assay/assay_20260414_122649/background_transformed_snapshot.png

Calibration:

/home/team8/z.avi_assay_tests/assayfinal/outputs/assay/assay_20260414_122649/calibration_snapshot.json

Duration (s): 9.8

Frames processed: 50

Tracking mode: stitch

Mask blobs as flies: True

Assay tubes: 4

Track fragments detected: 30

Unique threshold crossings: 4

0 s to 9.8 s comparison window

0 s to 9.8 s x-displacement summary
Columns 1-6 of 9 | Rows 1-4 of 4

assay_tube_index	flies_compared	mean_abs_dx_px	median_abs_dx_px	mean_signed_dx_px	comparison_target_s
1	6	15.553	5.113	-14.235	9.8
2	3	13.515	10.846	13.515	9.8
3	2	24.188	24.188	24.188	9.8
4	8	12.432	1.497	-9.372	9.8

0 s to 9.8 s x-displacement summary
Columns 7-9 of 9 | Rows 1-4 of 4

mean_abs_dx_mm	median_abs_dx_mm	mean_signed_dx_mm
1.503	0.494	-1.375
1.43	1.148	1.43
2.533	2.533	2.533
1.359	0.164	-1.024

Threshold crossing summary
Columns 1-5 of 5 | Rows 1-4 of 4

assay_tube_index	number_of_flies_detected	number_of_unique_threshold_crossings	fraction_crossing_by_10s	first_crossing_time_s
1	10	2	0.2	7.0
2	8	1	0.125	4.4
3	2	0	0.0	nan
4	10	1	0.1	3.4

Threshold crossing events
Columns 1-6 of 8 | Rows 1-4 of 4

unique_event_id	assay_tube_index	display_id	crossing_time_s	crossing_x_px	crossing_y_px
1	1	4	7.0	185.512	253.496
2	1	9	7.8	268.37	263.471
3	2	2	4.4	381.805	253.506
4	4	9	3.4	824.136	270.727

Threshold crossing events
Columns 7-8 of 8 | Rows 1-4 of 4

threshold_distance_px	threshold_distance_mm
501.0	66.912
501.0	66.912
498.0	68.103
500.0	68.143

Velocity summary
Columns 1-6 of 6 | Rows 1-4 of 4

assay_tube_index	number_of_flies_detected	mean_speed_mm_s	max_speed_mm_s	mean_vertical_velocity_mm_s	max_vertical_velocity_mm_s
1	10	1.466	16.717	0.73	16.368
2	8	1.764	9.395	-0.046	9.192
3	2	1.002	6.647	-0.595	1.775
4	10	0.953	13.291	-0.417	8.535